

Feature Guide: Research Lab

Health Transformation Review (HTR) Platform

What Is the Research Lab?

The Research Lab is HTR's suite of 21 interactive modeling and analysis tools. Each tool lets you input your own parameters and generate quantitative outputs — financial projections, population models, policy impact analyses, and more.

Entry point: /research-lab

The Lab is organized into six topic groups, each accessible from the hub page.

Lab Groups & Tools

1. Interoperability Lab — /research-lab/interoperability

Tab	Tool	What It Does
? tab=fhir	FHIR Interoperability Lab	Explore FHIR R4 data exchange, CMS interoperability rule compliance, API design for healthcare
? tab=risk	Risk Stratification Engine	Build and test clinical risk scoring models; explore data requirements and model validation

Best for: Health IT professionals, clinical informaticists, researchers studying interoperability

2. Payment Models Lab — /research-lab/payment-models

Tab	Tool	What It Does
?tab=apm-design	APM Design Lab	Design alternative payment model structures: MSSP, bundles, direct contracting, global budgets
?tab=apm-calc	Shared Savings Calculator	Project shared savings/losses under different benchmark methodologies and risk corridors
?tab=cea	CEA Calculator	Cost-effectiveness analysis: calculate ICER, model QALYs, apply standard willingness-to-pay thresholds
?tab=gb-transition	Global Budget Transition Modeler	Model a state's transition from fee-for-service to a global hospital budget

Best for: Health economists, actuaries, hospital finance teams, payment model consultants

3. Population & Equity Lab — /research-lab/population-equity

Tab	Tool	What It Does
? tab=population	Population Health Modeler	Build population models: chronic disease burden, risk stratification, preventive care impact, panel management
?tab=equity	Health Equity Studio	Analyze equity gaps by race, income, geography; explore SDOH drivers; model interventions

Best for: Population health managers, equity officers, public health researchers, clinicians

4. Policy & Quality Lab — /research-lab/policy-quality

Tab	Tool	What It Does
?tab=policy	Policy Simulator	Model the downstream impact of proposed legislation or rule changes on utilization, cost, and coverage
?tab=quality	Clinical Quality Optimizer	Optimize HEDIS, CMS Star, and VBC quality metrics; model protocol change impact
?tab=scorecard	Hospital Financial Stress Test	Model how policy or payment changes affect hospital margins
?tab=hta	HTA Studio	Health Technology Assessment following ICER, NICE, and HTI-2 methodologies
?tab=actuarial	Actuarial Lab	Capitation rate development, premium modeling, coverage cost projections
?tab=medicaid-wr	Work Requirements Calculator	Model Medicaid enrollment and disenrollment under work requirement policies
?tab=hr1-cliff	H.R. 1 Cliff Scenario	Analyze the financial cliff for states and providers if H.R. 1 Medicaid changes are enacted

Best for: Policy analysts, compliance officers, actuaries, hospital finance, researchers

5. Technology & AI Lab — /research-lab/technology-ai

Tab	Tool	What It Does
?tab=ai	AI Clinical Governance Lab	Build AI governance frameworks: FDA SaMD classification, bias testing, clinical validation checklists
?tab=digital	Digital Health Lab	Analyze digital health product pathways: reimbursement, regulatory clearance, ROI modeling

Best for: Health IT professionals, clinical informaticists, product managers, digital health investors

6. Knowledge Workspace — /research-lab/knowledge-workspace

Tab	Tool	What It Does
?tab=scorecard	Transformation Scorecard	Assess organizational transformation readiness across all six pillars
?tab=readiness	VBC Readiness Assessment	Detailed assessment of an organization's value-based care readiness
?tab=evidence	Evidence Library	Curated, searchable library of peer-reviewed research and policy literature
?tab=workforce	Workforce Modeler	Model healthcare workforce supply/demand, training pipelines, labor cost trends
?tab=leaderboard	Innovation Leaderboard	Benchmark innovation leaders across health systems, payers, and states
?tab=workspace	Research Workspace	Multi-tool research environment for extended analysis sessions

Best for: All users; the Evidence Library and Research Workspace are useful for every role

How to Use a Research Lab Tool

- 1. **Navigate** to the lab group URL and click the relevant tab
- 2. **Read the tool description** at the top — each tool explains its methodology
- 3. **Enter your parameters** — most tools have a form or input panel on the left

- 4. **Review outputs** — results appear as charts, tables, or narrative summaries
- 5. **Ask the AI** — the AI Analyst (right sidebar) can interpret results and answer questions
- 6. **Bookmark** the tool for future reference

Standalone Tools

Beyond the Research Lab, HTR has several standalone simulation tools:

Tool	URL	Use Case
HTR Transformation Readiness Simulator	/htr-simulator	Flagship org-level readiness assessment
Vermont Medicaid Eligibility Simulator	/medicaid-eligibility-simulator	Vermont Medicaid eligibility determination
Vermont Act 167 Simulator	/vermont-act-167/simulator	Act 167 scenario modeling
Vermont Act 68 Simulator	/vermont-act-68/simulator	Act 68 scenario modeling
CalAIM Simulator	/california-calaim/simulator	California CalAIM program modeling
CMS Rural Health Transformation Simulator	/dashboard/simulator	CMS rural health program scenarios
Impact Simulation	/impact-simulation	General healthcare intervention impact
Transformation Friction Index	/transformation-friction-index	Organizational friction analysis

Tips

- **Combine tools** — run the Policy Simulator, then validate financial impact with the Hospital Financial Stress Test
- **Ask the AI** to help you interpret outputs or suggest which tool to use for your question
- **Save results** — bookmark your current tool URL; outputs are session-specific
- **The Evidence Library** is your best starting point when you need research backing for your analysis

Last updated: May 2026 | Health Transformation Review